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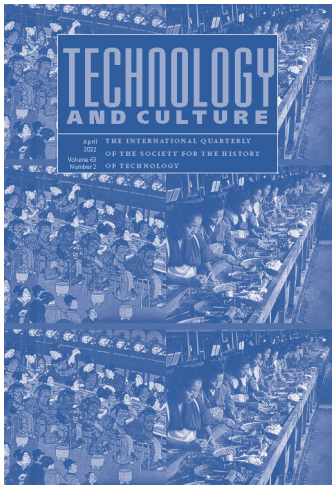
Energy Workarounds: Designing Coals for the Japanese Steel Industry, 1895–1911

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SPECIAL ISSUE

Energy Workarounds

Designing Coals for the Japanese Steel Industry, 1895–1911

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ABSTRACT: Japan's Meiji oligarchs put a premium on technologies that projected "civilization" and "modernity" and operated under the assumption that industrial technologies could be operationalized reasonably promptly. Their faith flew in the face of production experience. The case of metallurgical coke manufacturing offers an example of what happened when imported technological systems dead-ended on the factory floor. Examining the production records of a Meiji-era chemical start-up, this article brings to focus the scope and scale of the creative labor needed to make imported technologies work on the ground. In so doing, it showcases innovative forces that formed the fabric of Japan's early industrialization as a corrective to the much-criticized but resilient notion that the country's industrial takeoff was enabled largely by technology transfer and local appropriation. By highlighting the creativity involved in designing coal inputs, this article opens new perspectives on the history of coals in East Asia.

Introduction

In the second half of the nineteenth century, Japan underwent a tectonic political and social shift. It started as a relatively isolated agrarian economy in the twilight of the Tokugawa period (1600–1868) and became Asia's first industrial power over the course of the Meiji period (1868–1912). The news of Qing China's defeat in the first Opium War (1839–42) had caused shock waves in Tokugawa Japan, and it became clear that the country would need to harness the might of steam engines and gunboats if it wanted to maintain sovereignty and territorial integrity. Beginning in the 1840s, both the central government and domain lords embarked on military reforms that went hand in hand with acquiring military equipment, along with the know-how to operate it. This pursuit accelerated after the regime change of 1868. The following Meiji period saw ambitious, bold—

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and occasionally haphazard—waves of purchasing equipment, hiring foreign experts, and sending Japanese to train abroad. This pursuit of technologies that helped project "civilization" and "modernity" was more than a public relations operation; starting in the 1850s, Japan was submitted to unequal treaties regimes, and Japan's oligarchs felt compelled to project "civilization" and "progress" as part of the struggle to reverse the economic and political damage that these regimes had caused.[1] Ultimately, this strategy was a success, and scholars have credited Japan with avoiding the fate of Qing China by importing modern industrial technology to its shores and becoming Asia's first industrialized power.

Despite the key role of technology in this pivotal and turbulent moment in Japan's modern history, historians to date have devoted much of their attention to the political and social aspects of the Tokugawa–Meiji transition. The country's industrialization remained, for the most part, the province of labor and economic historians, studied within contexts of international economic development and comparative technology transfer. While there is a consensus that technology transfer was an important factor, the nuts and bolts of this transfer have to date remained largely unexplored, as economic historians have rarely discussed technology in detail. The resulting body of literature has contributed to the conception of Japan as a talented borrower and successful imitator of Western ideas and technologies.[2] At the same time, Japan's industrialization experience has been offered as a blueprint for the Cold War's other late modernizers.[3] The "Japanese model" was particularly potent in the 1970s and 1980s. Japan's spectacular economic recovery from the devastations of World War II was recent memory, and its high-speed growth (and trade deficit with the United States) had become front-page news. Japan's nineteenth-century industrial history thus became hostage to Cold War debates; the "Japanese miracle" was seen as a model for developing nations and influenced both sides of technical aid programs well into the 1990s.[4]

Nevertheless, the technologies of Meiji industrialization, at the core of Japan's success, remain empirically poorly understood. Historians of technology have shown that the nation and the engineering profession grew up together and that the overlap is not a coincidence.[5] Meiji nation-building was, in many ways, literally about building: reengineering the Shogunate into a parliamentary monarchy was a sizable political undertaking, but it also involved actual engineering projects. And yet we know far more about

1. Wittner, *Technology*.

2. Cox, *Culture of Copying in Japan*.

3. For recent examples: Hayashi, *Japanese Experience*; Minami et al., *Acquiring, Adapting and Developing Technologies*; Yoshihara, *Japanese Economic Development*.

4. Seely, "Historical Patterns," 15, 21.

5. Johnson, "Material Experiments"; Shen, "Taking to the Field"; Alder, *Engineering the Revolution*.

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Japan's modern myths than we know about the materiality and technologies (both industrial and quotidian) that went into this massive tectonic shift.[6] Despite available sources, there is remarkably little written about what it meant for Japan's engineers and technicians (key actors of Meiji industrialization) to deal with technologies designed and developed elsewhere.[7] Based on early successes such as the telegraph and railways, more than one Meiji oligarch assumed that existing industrial technologies could be easily deployed in Japan for considerable profit, albeit after some investment effort and engineering ingenuity. Yet rare glimpses from production archives show that much more was at work when Meiji engineers sought to install industrial systems and keep them running.

Looking critically at technology transfer as one of the key enablers of Meiji industrialization, this article aims to add nuance to two common assumptions. The first is the idea that industrial technology was "plug and play," that is to say, it could be transferred and become profitable after a modicum of tinkering. For example, Meiji entrepreneurs may have thought that the by-products of coke manufacturing could be easily and quickly commercialized: it seemed like making money out of (industrial) waste.[8] However, it turned out that substantial innovation and creative effort were needed to make ostensibly turnkey industrial solutions actually work on their arrival in Japan. The second assumption needing an update is that what has been labeled "technology transfer" by contemporary political and business actors was in fact a transfer of technology. A growing number of case studies that draw on careful examination of production archives show that the label "technology transfer" can hide a hugely creative albeit discreet process. Diverse imported technologies were inoperable on landing and required a range of creative strategies, including the redesigning of industrial inputs in unprecedented ways, in order to become operable. In spite of the scholarship done that proves otherwise, the idea that Japan industrialized not through innovation but by skilled copying remains remarkably resilient.[9]

Available archives tend to tell a different story from the one suggested by these two assumptions. The example of coke manufacturing in Osaka—indispensable for an industry as strategic as steel—demonstrates that successfully implementing foreign technology designs had relatively little to

6. Classics include: Gluck, *Japan's Modern Myths*; Fujitani, *Splendid Monarchy*.

7. For scholarship on engineers from a later period: Moore, *Constructing East Asia*; Mizuno, Moore, and DiMoia, *Engineering Asia*.

8. For similar attitudes to by-product placement in Republican China: Muscolino, "Energy and Enterprise."

9. Interestingly, the idea that the Japanese are dangerously good at technical espionage did not originate in Europe but dates back to sixteenth-century Korean perceptions of Japanese merchants. For details: Carré, *Avant la tempête*. This idea got a second wind at the turn of the twentieth century in the shape of the trope of Japan's deficient business morality: Hunter, "Deficient in Commercial Morality?"

do with imitation or mere tinkering with the imported equipment. It relied significantly more on the development of unprecedented industrial protocols along with the redesigning of inputs. This essentially meant doctoring raw materials to make them work around the limitations of imported technologies designed to use natural resources different from those in Japan. The difference in the quality of resources is striking: ovens had been designed with the assumption that coals coked would naturally contain between 16 and 22 percent volatile content, a cocktail of aromatic hydrocarbons found in coals. But Japanese coals averaged around 40 percent volatile content. This design assumption about the nature of coal deposits, invisible at the place of design, was a deal breaker on arrival in Japan. Thus, fueling the Japanese steel industry required creativity to work around the design assumptions of technology by effectively reengineering "close-enough" carbon inputs from very different mineral deposits.

Reasoning by analogy, the process at work in Osaka from 1896 to about 1911 resembles "carbo-forming." If terraforming refers to a hypothetical process of altering planets to replicate conditions on early Earth, carbo-forming describes the essence of a patented production protocol in Japan at the turn of the twentieth century. Its purpose was to coax Japanese coals to resemble the chemical structure of European coals. This process echoes contemporaneous North American practices of altering coal and mixing crude oil in pursuit of better performance.[10] However, the key difference was that—in Japan—mixing and doctoring were done not to improve the coal's performance in some absolute sense but to make Japanese coals fit for hardware designed with completely different coals in mind. In the process, Meiji engineering started to resemble haute couture carbon fuel designers. The records kept by the chief engineers at Osaka Chemical Works demonstrate that his team found it more effective to carbo-form Japanese bituminous coals to suit the Semet-Solvay coking ovens than to adapt the ovens themselves to Japanese coals. The technology transfer paradigm suggests that companies adapt imported technologies to match available natural resources, but in this case the reverse was true: engineers adapted local industrial inputs to fit the technology. Such a story showcases how Japanese engineers devised ways to make industrial technology work by actually producing—rather than simply harnessing—natural resources, albeit at great environmental cost and without claiming the "innovation credit."

The history of coke making in Japan is of interest to historians of technology for two reasons. In the aftermath of World War II, Japan emerged as the third-largest steel exporter in the world, ascending to the top place in 1973, then falling to second place by 2016.[11] As coke is essential for

10. Lucier, *Scientists and Swindlers*.

11. Yonekura, *Japanese Iron and Steel Industry*, 2; Global Steel Trade Monitor, "Steel Exports Report: Japan," March 2017, 1, https://legacy.trade.gov/steel/archives/index.asp

industrial-scale steel production, early accounts of coking in Japan are an important part of steel's global history. In addition, these attempts to produce coke contribute to broader debates on natural resources and technological innovation in the global industrialization story.

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Turning Noxious Fumes into Profit

In geological terms, coal comes from an intricate series of interactions between light, water, and carbon dioxide under various degrees of pressure over millions of years. Coalification is based on the death and decomposition of vegetation, whereby carbohydrates become compressed into stratified rock. For all their black lifelessness, coal strata are vast stores of energy-rich compounds in forms that can be exploited on extraction (or collection). Although habits of mind refer to coal in the singular, the rock is far from a monolith. The thickness of the coal beds and their carbon content, and therefore the quality of coal, vary greatly, depending on its deposition. The process in Asia, Europe, and eastern North America took place between 360 and 300 million years ago, during the geological period appropriately called the Carboniferous. Significant portions of the earth's surface were at the time covered in swamps and thick vegetation. After plants died and scavenging bacteria feasted on them, remnants of the plant matter decomposed in swamp waters. Subsequent generations of dead plants piled on top, compressing partially decayed vegetation into beds of peat. A range of paleo-environmental factors affected the thickness and spread of peat beds. Once swamps dried up and peat beds got buried by sediment, physical pressure became the main force of coalification, with the variation of geothermal conditions determining the quality of the coal sediment. Consequently, coal deposits varied across space.

In the course of the nineteenth century, coke—solid carbon fuel produced by processing coal—emerged as virtually indispensable in industrial-scale iron and steel production. In an age of railroads and battleships, steel production was as central to nation building in Japan as elsewhere.[12] One particularity of Japan's pursuit of industrial steel was the tension between steel's medieval and early modern history, on the one hand, and requirements of modern mass production, on the other. Steel was not a newcomer to Japan. Swords from the Kamakura period (1185–1333) show that —compared to their European counterparts—Japanese swordsmiths had an exceptional mastery of different kinds of steel and a superb understanding of how to work with them. Kamakura-period masters were capable of making swords with a soft core and a hard edge, meaning that a sword could bend without breaking while cutting fast and clean. In the 1610s, gunsmiths like Noda Kiyotaka demonstrated a sophisticated under-

12. Misa, *Nation of Steel*.

standing of the methods and merits of combining different steels. Noda's triple-folding method made strategic use of a combination of steels—steel smelted from iron sands locally, the imported steel made of iron ore, and gold-copper steel alloys—resulting in a remarkably durable product.[13] Beyond swords, Tokugawa period smiths produced and repaired guns. In terms of volume, arguably more iron and steel were used for repair and production of matchlock guns, which were in wide use among peasants, than for swords and armor for the samurai.[14] Yet despite this illustrious history, the steel used for railroads and battleships was a very different material. It required other raw materials and fuel and a very different production mode. Thus, the mastery of iron and steel used for swords had little bearing on the mastery of iron ore steel smelting for railroads. For all practical purposes, these were two different materials.

Under the threat of foreign encroachment, various Japanese domain lords experimented with new smelting protocols compatible with large-scale production of steel. Well into the Meiji period, this pursuit frustratingly failed to achieve their goals in large part because of the lack of a steady and sustainable energy source and a production protocol that could approximate consistent quality at reasonable cost. Despite the real and imagined mastery of artisanal steel, failure and frustration marked the Meiji pursuit of mastery of industrial steel. As Japan prevailed in the Sino-Japanese War (1895) and was headed into a confrontation with Russia (1904–5), the issues around coal and steel acquired new urgency. Coke turned out to be a key bottleneck.

Manufacturing coke involves heating coals in a retort (coke oven) in the absence of air and driving out moisture and volatile matter (impurities). Volatile content or volatile matter greatly determines the properties of coals and depends on the geological age, deposition, and sedimentation of the coal beds. For the purpose of coke making, the higher the proportion of impurities, the larger the quantity of emissions in the coking process. In other words, coke is what comes out after coals are baked in an oven and impurities such as noxious gases and toxic effluents are eliminated. The chemical composition of the emissions depends on the specific content of the volatile cocktail being coked.

Standard 1880s coking technology was a beehive method, where coals were fed into a row of retorts for 48–72 hours. There was no provision for recouping the fumes and effluvia from the manufacturing process. Yet in the late nineteenth century, one industry's waste product became another industry's vital feedstock. The noxious fumes and obnoxious odors contained chemical components that could be identified, recovered, and commercialized. For example, ammonia was indispensable for the global dye,

13. Sugawa, *Nihon no hinawaju*; Howell, "Social Life of Firearms."

14. Manako Tanaka, "Understanding Materials and Production Process of Japanese Matchlock Guns Using Non-destructive Analysis" (unpublished working paper, 2012).

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soap, glass, and chemical industries. Railroads used coal tars for wood protection, coal gas was used for illumination, and benzene found multiple takers, including the food industry (for instant coffee).

Unsurprisingly, the chemical industry was key in the shift from beehive to by-product coking. A major player was Solvay & Cie., one of the success stories of Europe's chemical industry. The company was established in Charleroi near Brussels in 1863 by Ernest Solvay, an ambitious and self-taught chemical engineer. His success and subsequent fortune came from patenting the process for manufacturing soda ash, an indispensable component in glass production, water treatment, and soap making. His method for making soda ash was more economical than the then-common Leblanc method, and the company grew accordingly. Solvay's principal drive to develop this by-product technology was not his interest in coke—ostensibly the main product of coking—but the need for a stable source of ammonia, a by-product of coking but a vital ingredient for Solvay's soda ash production business. Early on, Solvay soda ash plants obtained their ammonia from surrounding gas plants, an arrangement that occasionally stressed supplies because of transportation difficulties.[15] With ever-expanding production, Solvay could ill afford an interrupted supply. Sizable capital investments and an increasing need to control the ammonia supply both motivated and enabled him to invest further in a method to recover by-products of coking and turn them into profitable commodities. Thus, the line between "main product" and "by-product" was blurred, as were the lines between "industrial waste" and "commercial products."[16] Furthermore, the route from waste to by-product was not a one-way street. Whether or not a by-product became waste depended not simply on its known uses but also on current market demand. If there was insufficient demand or storage was an issue, by-products could quickly become waste, contributing to the industry's toxic legacy.[17]

In 1876, Solvay commissioned Louis Semet to develop a coking oven that could retrieve by-products. As a family-owned and family-run company (which it remains), family ties were important for Solvay & Cie. Semet was not only Ernest Solvay's cousin but also his brother-in-law. Not only were they related, but Semet was also well positioned and well qualified for the challenge, having started his business career in lighting gas production, which relied on coal combustion emissions.[18] Nonetheless, the prospect of developing a coking protocol that would allow the recovery of emissions at a reasonable cost in the late 1870s seemed bleak. A number of

15. Bertrams, Coupain, and Homburg, *Solvay*, 59.

16. Sarah Teasley, "Waste Wanted? Sawdust, Warped Boards and Other 'Waste' Materials in Postwar Japanese Manufacturing," paper presented at the 2018 Association of Asian Studies Annual Meeting.

17. Tarr, "Toxic Legacy."

18. Four à Coke Semet-Solvay, *Four à Coke Semet-Solvay*, 63.

experiments took place across Europe in the 1860s but failed for several reasons: either they made coke that was unsuitable for iron and steel production (sub-metallurgical grade), or recovery costs were too high, making the by-products commercially unviable.[19] Yet after several years of experimentation, in early 1882 Semet built a test plant with six ovens at Mons, Belgium (40 kilometers west of the original Solvay plant at Charleroi). He built a second battery of twenty-five ovens in 1886 at the Havré coal mine just outside Mons. By this point, the protocol was economically efficient enough to allow Solvay to develop a particularly effective business model. Typically, the company offered to build ovens at its own cost in a customer's facility. The client retained all the coke produced, while Solvay recouped its investment over time and made a profit by using and selling all the by-products. This arrangement benefited clients with high-energy needs that were unwilling or unable to invest financial capital in building and operating a successful coking plant. Business was brisk, and it soon spread across the Atlantic. In 1892, the first Solvay ovens were built in the United States, in Syracuse, New York, followed by installations in Pennsylvania, Alabama, Kentucky, Ohio, Michigan, West Virginia, Massachusetts, and Canada. In Europe, Solvay expanded its operations to Germany, Italy, and Britain.

Coinciding with Solvay's by-product ventures outside Belgium, on the other side of the globe Osaka was emerging as Japan's industrial metropolis. Although Tokyo would later become the nation's premier industrial site, Osaka initially overshadowed Tokyo in terms of production output and labor force.[20] The concentration of the chemical industry in Osaka is particularly noteworthy. In 1902, Osaka's chemical industry had twice as many plants and workers as Tokyo.[21] Equally important for understanding the demand for coke is the fact that a quarter of Japan's machine-tool factories were clustered in Osaka and its suburbs, employing almost a fifth of the nation's metal workers.[22] One of Osaka's chemical plants originated when Osaka entrepreneur Toyama Shūzō 富山修造 visited organic chemistry professor Shimomura Kōtarō 下村孝太郎 in Kyoto in the winter of 1895–96.[23] Both men are relatively obscure today, and not much is known about the nature of the discussion, but Shimomura described the event as a "fortuitous meeting." In all likelihood, it was because Shimomura was convincing enough on the potential profits from commercially valuable by-products of coke manufacturing.[24] Later in 1896, Shimomura quit

19. Burn, *Economic History of Steelmaking*, 204.

20. Sawai, *Kindai Ôsaka no Sangyô Hatten*; Abe, *Kindai Ôsaka keizai shi*.

21. Industrial census data in "Kôjo tsûran."

22. Sawai, *Kindai Ôsaka no Sangyô Hatten*.

23. Names follow Japanese convention: the family name precedes given name.

24. "Shimomura Kōtarō kôgyô rireki" [Industrial employment history of Shimomura Kōtarō], 1925, Folder 13, RG: Y-SK-G, DUA.

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teaching and took a job in the chemical start-up. Toyama sent him to the United States and Europe in search of "the best by-product coke ovens."[25] By the time the professor became chief engineer at Osaka Chemical Works, he was tasked with setting up a coke manufacturing facility. He first encountered Semet-Solvay by-product coking installations in New York State before visiting Solvay at its Belgian headquarters in 1897.

Industrializing in a Time of Urgency

As fine-grained studies of Meiji industrialization "successes and failures" reveal, buying and installing imported machinery was a challenge, but running a production site and turning a profit for sustained periods of time was an exercise in juggling challenges on thin ice. As elsewhere, the curriculum for Meiji engineering students at Tokyo Imperial University was a staple of standard textbooks distilling experiences and solving problems of contemporary industrial production in Europe and North America. Foremen and technicians often considered these engineering graduates inept in a production setting.[26] They were known for arriving at the factory in their Western suits and white gloves, projecting professional elite status. While useless, elite engineers being unable to solve production problems is far from a uniquely Japanese stereotype, its existence in Japan points to an often-neglected reality: much of existing textbooks and reference literature had little guidance applicable to common Meiji industrial scenarios. For example, oligarchs and many bureaucrats assumed that technology was universal and thus transferable. Engineers, however, were confronted with systems and protocols built on the assumed availability of a particular raw material that was difficult to work around but invisible to a non-specialist's eye. Tasking engineers with making imported technologies work in Japan often came down to asking them to make things work in ways they were not quite supposed to work or to devise solutions to problems of great urgency in Japan but not of great interest in the European context. What in discourse were glitches to be solved and details to be tweaked were sometimes assumptions hardcoded into industrial technologies in a way that transformed "details" into "showstoppers."

In the case of coking this would prove to be very true, and Shimomura Kōtarō walked into just one such Meiji industrial problem. Before turning to industry, Shimomura was looking into a career in teaching chemistry. He held a Ph.D. in organic chemistry from Johns Hopkins and started his professional life as the president of the Harris School of Science at the University of Dōshisha in Kyoto (see figure 1). For various reasons, he did not see himself foremost as a man of industry at first, but a conflict with

25. Letter Shimomura to Worcester Polytechnic Institute's President Mendenthall, January 26, 1899, Folder 51, RG: Y-SK-G, DUA.

26. Pauer, "Search for (Social) Identity."



FIG. 1 Shimomura Kōtarō (1861–1937). Graduate of Dôshisha University and Worcester Polytechnic Institute, he obtained his Ph.D. in organic chemistry under Professor Ira Remsen at Johns Hopkins University. He was the founding president of the Harris School of Science, the first college-level scientific school in Kyoto, established in 1890. (Courtesy of DUA.)

the university administration drove him to resign and pursue opportunities in Osaka's budding chemical industry. By 1897, as chief engineer at Osaka Chemical Works, he embraced the challenge of building the company from scratch. He did not belong to the elite group of engineers from Japan's Imperial College and had not studied in Germany, a key rite of passage for top engineering echelons of his generation. Independent of whether Tokyo University engineers considered him a peer, Shimomura was a quintessential Meiji engineer in an operational, rather than institutional pedigree, sense. In August 1897, recently back from Europe, Shimomura placed two orders: one for a coal mixing, washing, and grinding plant from Hardy Patent Pick Co. in Sheffield, and one for sixteen Semet-Solvay ovens from Brussels. Ordering fewer than the standard number of ovens (twenty-five) for a medium coking plant indicated a cautious start. Yet the costly Semet-Solvay ovens were signs of ambition and chutzpah. Before operations could be successful and profitable, he would have to overcome the major obstacle of coking Japanese coals in Belgian ovens.

While in steam power generation different coals could be substituted for one another, in metallurgical coke manufacturing small differences in coal quality were a big problem. The point of reference for Semet-Solvay ovens, in terms of assumed coal quality, were mostly Belgian, French, and German coals with an average of 15–18 percent volatile matter. According to company records from about the same period, no Solvay oven coked coals with more than 22 percent volatile matter.[27]

The crux of Shimomura's challenge lay in the substantial differences between the coals Solvay ovens were designed to coke and those he had at his disposal. Practically all of Japan's coals were high in volatile matter content, between 38 and 40 percent. The higher the volatile matter in coals

27. Four à Coke Semet-Solvay, *Four à Coke Semet-Solvay*, 63.

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coked, the lower the quality of coke obtained and the less suitable it was for iron and steel production. Even a small percentage of impurities significantly influenced the quality of coke produced. As Japanese coals had on average twice as many impurities as the coals the ovens were intended for, this posed a considerable technical challenge that was not addressed in the technical literature of the day, as it was a problem that European and North American coke manufacturers did not have.[28]

Though it is common to refer to coal in the singular, because of the wide varieties in chemical composition, it is more appropriate to talk of coals in the plural. The quality of coal is not simply on a spectrum from poor to excellent. Coals must be matched to specific uses. As Christopher F. Jones aptly shows, there was nothing inherently and obviously superior in anthracite coals when they appeared on the market. Most mid-Atlantic residents initially ignored anthracite coals, and when unable to sell or devise a use for them, merchants dumped the "rock that would not burn" in the streets for use as gravel.[29] Even once people learned how to handle anthracite, it could not be used for manufacturing lighting gas. For that purpose, bituminous coals—ostensibly of poorer quality owing to lower carbon content—were a perfect fit. Their impurities were a blessing, whereas high carbon content was a curse. It is therefore more accurate to speak of coals and bear in mind that there were no good or bad coals, only more or less appropriate ones. Although cast in the interwar period as resource poor, Japan was at the same time both an importer and exporter of coal.[30] The struggle to manufacture metallurgical coke at Osaka was about knowing how to manipulate the coals available to achieve satisfactory results in coke ovens that could essentially not be changed. Hence, the key to successfully running a coking plant in Osaka was not just purchasing, assembling, and adapting Belgian and British equipment to suit local conditions. Equally, it was not Japan's self-proclaimed poverty in mineral energy resources that made coke manufacturing such a challenge. It was about matching the gap of affordances. Domestic coals were ill-suited for coking using the available technology (even top-of-the-line Semet-Solvay ovens). Existing coking technologies simply could not deal with the quantity and range of volatile elements in most Japanese coals. There was a mismatch between what kinds of coals were specified and what kinds were accessible. Before the ovens could produce the desired result—hard coke suitable for smelting—the technicians working with Shimomura had to reimagine the coking process and change the definition of what constituted a "coking coal" (see figure 2).

That manufacturing coke from domestic coals would be a problem in

28. For a reference work available to Shimomura: Fulton, *Coke*.

29. Jones, *Routes of Power*, 27–49.

30. For coal exporting: Phipps, *Empires on the Waterfront*; Dusinberre, *Hard Times in the Hometown*. For interwar anxieties about natural resource scarcity that shaped historical analysis of Japan's industrialization: Dinmore, "Small Island Nation."

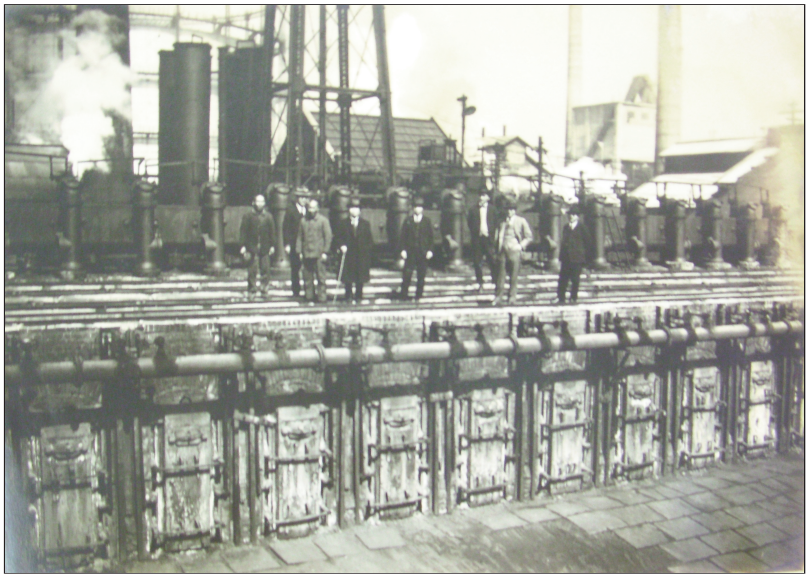


FIG. 2 The first battery of byproduct coke ovens installed in Japan at Osaka Chemical Works. Photograph taken on March 9, 1931, days before their dismantlement after more than thirty years of active service. (Courtesy of DUA.)

Japan was no surprise. Starting in the 1850s, most smelting experiments encountered issues with coke. Preceding Osaka Chemical Works' entry into coke manufacturing in 1898 was a five-decade history of failure, notably (but not only) at Kamaishi Iron Works.[31] The impossibility of turning Japanese coals into metallurgical-grade coke was a given in international and domestic professional circles. With this in mind, the purchase of Semet-Solvay coke ovens could be construed as ignorance of earlier attempts or an inability to draw the right conclusions from past experience. Yet it would be misleading to write this off as a poor technological choice. Breaking new ground and pushing the boundaries of what was considered possible drove the 1890s-era efforts to get the steel industry in Japan up and running. The Osaka coking venture was no different. Added to the omnipresent drive to catch up with the industrialized world was the urgency of putting steel to use in transportation, armaments, and the machine-tool industry during and after the Sino-Japanese War.[32]

Victory in the war was taken as a sign of Japan's rising political and industrial muscle. Less often remembered is that Japan won rather unexpectedly, not thanks to its superior naval power but due to a toxic internal conflict in the Qing navy. Subsequent visual depictions of a confrontation between Qing junks and Meiji iron ships are misleading. Both navies were

31. Wittner, *Technology*.
32. Sawai, *Kindai Osaka no kôgyô kyôiku*; Suzuki, *Meiji Kikai Kôgyô*.

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equipped with foreign-built ironclad ships. Moreover, the Qing navy had a few high-tech surprises up its sleeve. Mid-battle, the Japanese navy experienced the hard way that not all ironclad ships were made equal. During an encounter, they realized that the Chinese Northern fleet's German-built Zhenyuan battleship (1884–1914) was unaffected by shells and thus practically immune to artillery attacks. Still, through a combination of good fortune and unconventional tactics, Japan won the war.

Consequently, under the provisions of the Treaty of Shimonoseki that ended the war (known in China as the Treaty of Maguan), Qing China had to pay Japan war reparations totaling 200 million taels of silver (7,400 tons). While politicians celebrated the victory, the Japanese navy's engineers and technicians were more cautious. They requested that the Chinese steel-plated flagship be delivered as a prize of war. It was promptly docked at the naval arsenal in Kure (Japan), where the Stettiner Vulcan–produced ship was taken apart for examination. At the time, Japan was nowhere near having a working steel mill. The Sino-Japanese War experience added urgency, and the colossal indemnities (2.3 times Japan's annual budget) provided the capital for a new, more ambitious government initiative in iron and steel production.[33] While construction of the Imperial Yawata Steel Works was underway on the southern island of Kyūshu, Shimomura was installing Semet-Solvay ovens at Osaka on the southwest island of Honshu.

Doctoring Japanese Coals to Fit Belgian Machines

How exactly do you redesign and manufacture Japanese coals to fit imported ovens? The mix of anxiety and urgency typical of Meiji industrial ventures produced particular attitudes toward the lessons of past failures, standard production protocols, and the conventional order of operations. This meant that when European colleagues confidently stated that his project was a folly and it was not how things were done, Shimomura discretely disagreed, operating under the unstated assumption that the reigning givens of the industrialization were there to be rewritten. For Shimomura, producing metallurgical-grade coke from local coals was simply a matter of time, and success was a matter of degree. Incremental innovation was the stuff of Meiji innovation. If at first the coke produced in Osaka was not suitable for smelting, not all was lost; it could be sold as domestic fuel. In the meantime, recouped and commercially viable by-products could generate income.[34] This bought time to experiment and explore how to redesign the firm's coking protocols.

One of the first coking conventions that Shimomura disregarded was

33. Yonekura, *Japanese Iron and Steel*, 37.

34. Letter Shimomura to Thomas C. Mendenhall, Worcester, U.S.A., January 26, 1899, Folder 51, RG: Y-SK-6, DUA.

the typical order of operation in preparing coals for coking. In June 1897, while drawing up specifications for the coking plant in Osaka, Shimomura ran the idea of reverse order of operation by Cecil Walton, an engineer at Hardy Patent Pick Co. in Sheffield. Coals were usually first washed and then ground before loading in ovens. Shimomura asked that the blueprints (which were to be sent with the equipment to help with the installation) project the reverse-order scenario: grind the coal first and wash after grinding. The idea caused a substantial stir. At first, the staff at Sheffield responded with a reserved "we do not recommend that coal should be ground before washing [as] it is usual here & on the continent to grind after washing."[35] But Shimomura insisted that the installation plans reflect his intention. The tone of the correspondence from Sheffield ramped up, and three months later Walton vehemently protested, stating it was "suicidal to grind this material before washing." If Shimomura "wish[ed] to be successful," Walton warned, he "must wash first and grind afterwards." Ultimately, Sheffield engineers refused to send any installation plans per Shimomura's specification, a de facto release of responsibility for performance. When the equipment arrived in Japan the same year, Shimomura set up the plant without a plan and contrary to the supplier's specification.[36]

There were good reasons for the advice sent from Sheffield. Washing before grinding meant that different shapes and sizes of coal lumps allowed a freer passage of water than what was possible when coal was ground to small pieces of about the same size. Reasoning by analogy, the difference is similar to washing rocks versus washing fine gravel. Washing coal after grinding required more time, labor, and water and caused more pollution. Shimomura was neither ignorant nor unreasonable. He knew full well the advantages of the conventional preparation protocols. Yet sidestepping them made perfect sense within the constraints of his operation. In Osaka, grinding coal first and washing it later meant physically disrupting the coal's structure. Having extensively tested different mixtures of Japanese coals, Shimomura knew that no method of impurity elimination was to be neglected for the Japanese coal—no matter how outlandish it sounded in Sheffield. Washing second meant that partially dislodged or freed impurities could be washed away before coking. Osaka Chemical Works therefore aimed to use the resources at hand (coals, water, and labor) creatively to contribute to building the archipelago's steel industry.

The second convention Shimomura disregarded was that on coal sourcing. Since medieval times, the Osaka-Kobe-Kyoto region was sourcing coals from west Japan, notably from the island of Kyūshu. Coals from recently opened mines in Hokkaido and in the northeast island of Honshu

35. Letters Hardy Patent Pick Co., Sheffield, to Shimomura, July 9, 1897, Folder 53, RG: Y-SK-6, DUA.

36. Letters Hardy Patent Pick Co. to Shimomura, September 1 and 6, 1898, Folder 53, RG: Y-SK-6, DUA.

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(Jōban) were destined for the increasingly coal-hungry Tokyo-Yokohama region. Instead of following a conventional sourcing strategy, Shimomura's test results reveal a wide-net and systematic approach to coal assaying. Since the start of operations in November 1898, Shimomura had ordered sample shipments of different coals from all over Japan, as far north as Yubari (Hokkaido) and as far south as Nagasaki (Kyūshu). "Every day new coals come to our hand," he wrote to a fellow engineer in China, "and different mixtures are being daily tested in the ovens."[37] Eighteen coal-producing sites in Japan were short-listed and tested for eight parameters, with Cardiff coal as a point of comparison. Shimomura proceeded to go beyond caloric value and overall percentage of volatile matter as indicators of a coal's suitability for coking. Systematic mixing based on parameter variation eventually yielded a key insight: much depended on the exact breakdown of the content of aromatic hydrocarbons making up the overall volatile content segment. After over a year of research, it became clear that if, within the 40 percent volatile matter mix, ash represented only 20 percent and the sulfur content was under 2 percent, Japanese coals could be nudged into yielding something that approximated blast furnace coke. The key piece of the puzzle was in the details of the chemical composition of the volatile matter rather than in its volume. Consequently, Shimomura had to find Japanese coals that, regardless of their composition, had tolerable ash and sulfur levels. After several months of further testing, by January 1899 Shimomura had identified four coals that could match the specification.

To confirm his findings, he sent samples of two tons each to be coked in Belgium's Solvay facilities. For obvious reasons, he gave no indication of origin and only numbered the samples. When the results came back, Shimomura noted that Yubari (Hokkaido) coals "sent to Belgium . . . gave a good coke." They had the additional advantage of sufficiently low sulfur content, acceptable ash levels, and—importantly—being available in large quantities from newly developing mines in Japan's northern territories. The Yubari coal mine on the northern island of Hokkaido was opened in 1889, twenty years after the Meiji government integrated the island into Japan's national territory, making it a laboratory of settler colonialism and extractive industries. In 1892, Yubari had a rail connection to the port of Muroan, and by the time Osaka Chemical Works sought a steady supply of coals with strict specifications, freight costs were reasonable enough for the company to acquire a share of the mine and thus secure supply.

The next challenge was a riddle: coal from the same batch that produced good coke in Belgium when sent for independent confirmation tests, produced splintered edge coke unfit for blast furnaces when coked at Osaka. Same coals, same ovens, different results in Belgium and in Japan.

37. Letter Shimomura to Chief Engineer, The Iron & Steel Works, China, January 18, 1899, Folder 51, RG: Y-SK-6, DUA.

Technically, there was an easy fix for the Osaka splinter problem that made coke unfit for blast-furnace use: adding imported anthracite to the mix of coals put into the ovens. Yet this was not a satisfactory solution because the enterprise sought to rely entirely on domestic resources. In 1898, Shimomura continued testing different mixtures and meticulously documenting results. Evidence of this intense testing campaign can be found in the amount of correspondence requesting urgent advice, including on where to get a machine that could thoroughly mix "several proportions of coal . . . proportions to be regulated at will."[38] Reading between the lines of his letters to Brussels reveals that Shimomura realized that mixing several kinds of coals in particular proportions would fix the problem. However, adding imported coals to the mix was a deal breaker for Shimomura's definition of success because self-sufficiency, meaning local sourcing of coals, was paramount. As a result, he spent most of 1899 testing different kinds of Japanese coals and figuring out which mixtures would eliminate the splinters. The volume of testing and assaying was growing by the day and drove him to desperately and urgently acquire a supplementary muffle furnace—a device for determining the ash content in coals. "We have so much analysis to do every day," Shimomura wrote, "that one determination at a time takes too long. . . . If you can, please send it as soon as possible."[39]

After two years of falling short of his goal, Shimomura disregarded yet another convention. To date, coking was a one-off process: coals were fed into the ovens once, and what came out after firing was considered the final product. At Osaka Chemical Works, coking became a two-step operation. First, a mix of Japanese bituminous high-volatile coals was coked at 400°C to 600°C, and the result was called X-Coal or semi-coke, with 15–20 percent volatile content. Second, X-Coal was mixed with high-volatile coal in a proportion ranging from 15 to 40 percent and then fed into a retort again. By doing this, Shimomura succeeded in designing a boutique Japanese coal mix that worked in Solvay ovens. After five years of yielding poor-quality coke, by 1904 Osaka Chemical Works successfully produced metallurgical-grade coke from local coals comparable to the coke from imported European coals. Shimomura's procedure enabled coke manufacturing from local coals, but with a steep environmental cost. In sum, the use of Japanese coal meant high tolerance for poor yields, much labor, and much environmental damage in exchange for developing a technology that could operate exclusively with fossil resources of the archipelago.

38. Letter Osaka Seimi Works Co. to Minami Shinden, Osaka, November 24, 1898, Folder 51, RG: Y-SK-G, DUA.

39. Letter Shimomura Kōtarō, Kyoto, to Solvay Co., Brussels, May 8, 1899, Folder 51, RG: Y-SK-G, DUA.

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While Shimomura was busy doctoring coals, the Imperial Yawata Steel Works was solemnly inaugurated with the completion of blast furnace no. 1 in November 1901. Before long, it became obvious that maintaining the blast furnaces, Bessemer converter, and open-hearth furnaces would be no easy feat. By February 1902, production encountered serious difficulties, and a few months later all activities at Yawata were halted. Japanese engineers agreed that the poor quality of the coke was the key obstacle to advancing production. When not enough coke was available or when coke was of sub-metallurgical grade, lumps of heated iron hardened on the walls of the blast furnace instead of flowing out in a molten state. This completely interrupted production, cooled down the oven, and meant that the blocked pipes and lower parts of the furnace had to be replaced—a long and costly process.

After production at Yatawa came to a halt in 1902, General Nakamura Yūjirō was appointed as director, signaling that the Meiji government was taking direct control of operations. Nakamura's tenure was marked by a series of drastic measures, such as the suspension of all attempts to blow in the blast furnace and the dismissal of most employees, including top management like Chief Engineer Ōshima Michitarō and W. Neuhauser, one of the two remaining German foremen. Operations resumed in April 1904 but were halted after just two weeks of trials, and further attempts to make blast furnace no. 1 work were abandoned. By that point, Japan was at war with Russia, and Japanese engineers had built a second blast furnace. Falling into the high-tech trap and focusing most of their energy on blast furnaces, Japanese engineers failed to pay much attention to technologies—such as coking—that they considered peripheral. Consequently, in-house coking operation continued performing poorly. Fearing that without good coke the second blast furnace would share the same fate as the first, a delegation of Yawata officials headed by General Nakamura visited Osaka Chemical Works. Shimomura was convincing in explaining that Osaka Chemical Works may have the solution for Yawata's coke problem. Nakamura was keen to get Shimomura's help but at the same time wary of revealing the full extent of the problem at Yawata or bringing him fully into the Yawata problem-solving team. Shimomura was different from most engineers Nakamura worked with at Yawata, who were graduates from the University of Tokyo's Engineering Faculty. Yawata's top engineers had a clear sense of their elite status, exhibited strong loyalty to their alma mater, and were convinced of its preeminence. Hence, it was hard for them to work under the guidance of a chemistry professor turned engineer who had not studied in Germany like them but had strong U.S. connections. On top of that, Shimomura was a practicing Protestant. In the late-Meiji mindset, Japanese Christians were considered incapable of being

fully patriotic; their loyalties were suspect, as they bowed to Jesus instead of the emperor and had perhaps too-close ties with foreigners. Inviting such a man into the country's major military industrial facility was not a step to be taken lightly, but desperate times required unprecedented measures. Shimomura's expertise and Yawata's dire situation finally outweighed suspicion and egos. In 1905, Shimomura was asked to help order 75 of the 150 Semet-Solvay coke ovens planned for Yawata. The following year, he made several trips to Kyūshu to install the ovens and train the engineers and foremen. The 75 ovens were installed in November 1906, and the rest by 1909. As noted above, the key element in coking local coal and getting metallurgical-grade coke was not learning to use the ovens but mastering the preparation and mixing of the coal fed into them. With Shimomura's procedures in place and the Semet-Solvay ovens operating, Yawata finally obtained the coke it needed for efficient operations. As a result, soon after the 150-oven coking plant became operational, Yawata registered its first profits in 1911.

Conclusion

This article contributes to three ongoing debates in the history of technology. First, the coking in an Osaka start-up in the 1890s is a source of materials and experiences that can generate a new conceptualization of the energy-industry nexus. Despite considerable internal diversity, Asia's energy history shares a common implicit understanding that coal is a geological artifact that a country does or does not have, had to survey to discover and mine, and was able (or not) to transport efficiently.[40] Despite much work questioning coal as a key factor of industrialization in Europe and elsewhere, the early industrial development in East Asia is concerned with access to coal. The case of early coke manufacturing in Japan invites a rethinking of the importance of access to coal. To make steel in Japan as in China, a steady supply of coal at a reasonable cost was important but not paramount. In fact, in the case of iron foundries and steel mills, access alone was meaningless unless there was a quality match between a particular use and a particular kind of coal. This echoes the work of Nadège Sougy on French coals, which shows that the focus on quantity is a poor analytical tool for explaining not only the French-British divergence but also the broader role of energy in industrialization processes.[41] Thinking of coal in the plural is the first step in a shift of conceptualizing of coal from that of a natural resource to that of a manufactured product. Osaka Chemical Works' early coking efforts demonstrate that industrial coals are not purely a geological lottery but an energy product made to specification.

40. Pomeranz, *Great Divergence*; Shen, *Unearthing the Nation*; Wu, *Empires of Coal*; Muscolino, "Energy and Enterprise."

41. Sougy, *Les Charbons*.

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Having plurality of coals in mind ushers in new avenues for understanding the energy-industry nexus in East Asia. This reconceptualization will manifest differently in different language contexts. In English and French, for example, it presents itself in the form of referring to coals in the plural. In Japanese, as nouns do not have distinct plural forms, it is impossible to pluralize. Yet this does not preclude the emergence of analytical and language tools that adequately communicate the plurality of coals and what that means for an industrial setting.[42]

Second, beyond a more nuanced understanding of Japan's early industry, reconceptualizing coals as products, not simply a natural resource, contributes to the critique of diffusionist logic in the history of technology. Powerfully criticized as the framework for understanding how technology circulates in the European context, it remains remarkably alive and well in the treatment of non-European cases in general and Japan in particular. Based on production archives dating from the late 1800s, this article adds nuance to our understanding of technology transfer and contributes to new research (and thinking) about how technology travels. It provides an empirically grounded critique of the contemporary assumption that industrial technologies travel well and are ultimately highly effective. Here, the Japanese story told in this article echoes the critique of the "imported magic" trope in Latin America.[43] The idea that Meiji industrialization was built out of technologies originating elsewhere is still a common trope in Meiji entrepreneurs' biographies and official company histories (*shashi* 社史). In principle, there is a consensus that little technology travels without some thinking and adaptation at the point of delivery. In practice, most accounts of how Meiji Japan industrialized under-analyze the creative labor involved in making imported technologies work or working around them. In contrast, the case of coke manufacturing in Osaka, Asia's preeminent early industrial metropolis, demonstrates the remarkable amounts of energy and creativity poured into working around imported technology. The assumptions built into their original design could be biases against their local acclamation.[44] Pointing out the limitations of diffusion and transfer narratives about Japan is all the more important because history of technology is going through a "global turn." Framing the modern history of technology outside Europe and North America as "imported magic" overlooks the ways different conceptualizations of technology were developed and deployed by historical actors in different corners of the world.[45]

42. For an example of work attentive to the importance of raw material diversity in chemical industry—in particular related to common salt, lime, and sulfur in alkali-acid manufacture: Kamatani, *Nihon kindai kagaku kōgyō no seiritsu*, 323–35.

43. Medina, da Costa Marques, and Holmes, *Beyond Imported Magic*.

44. For exceptions: Tetsurô, *Nihon kindai gijutsu no keisei*; Hunter, "Reviving the Kansai Cotton Industry"; and Choi, "Another Spinning Innovation."

45. Medina, da Costa Marques, and Holmes, *Beyond Imported Magic*, 2–3.

Third, this article also begs the question about the analytical limits of adaptation when moving technology across distances. At what point do tinkering and adaptation stop being "transfer" and start being "innovation"? In other words, under what conditions can and should "appropriation" assume the epistemic dignity of "innovation?" Does the projection of broad power dynamics cast a longer-than-warranted shadow over the process that would otherwise come together under different terms if it happened elsewhere? The nature of what was going on at Osaka Chemical Works with coke manufacturing stretches the notions of appropriation and technology transfer to a breaking point. Working around and against imported technologies required much time and effort; the labor was incremental and innovative, despite historical actors' claims. It is impossible to prove that the same work that was done by Shimomura in Osaka would have been considered differently had the technology traveled the other way around. What is clear is that detailed mapping of the 1890s trial-and-error metallurgical-grade coke manufacturing in Osaka confirms that even a very nuanced understanding of the adaptation/tinkering/appropriation framework is too blunt of a tool to restore to the record the creativity and innovativeness that are the neglected connecting tissue of Japan's early industrialization.

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